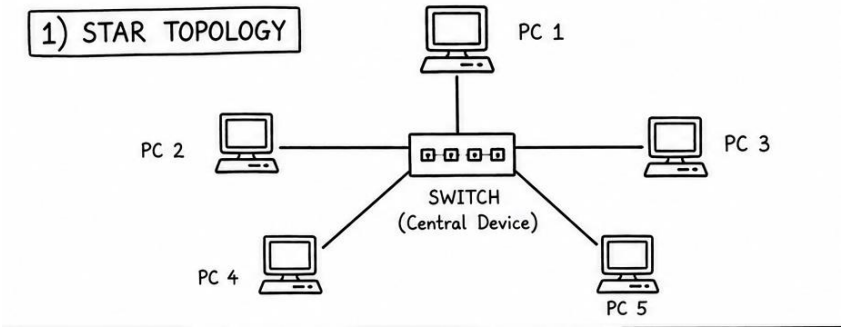
Network Topologies :-

Tasks :-

1. Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

Ans. :-

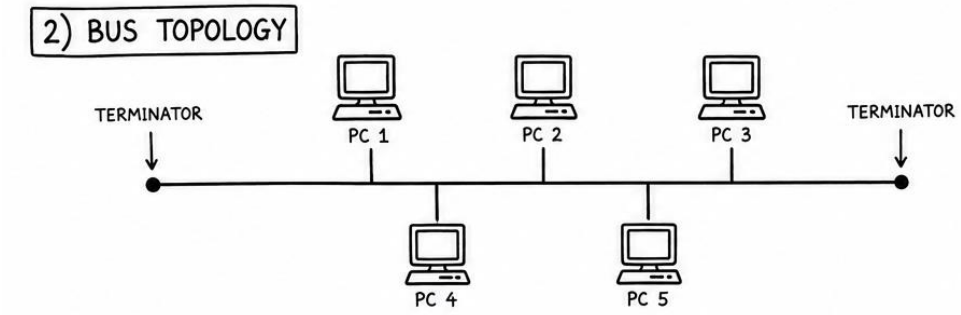
Star Topologies :-



WhatsApp uses central servers to send messages between users. Like a star topology, every device communicates through a central point.

Star Topology:- Best for homes, schools, and offices because it is reliable and easy to manage.

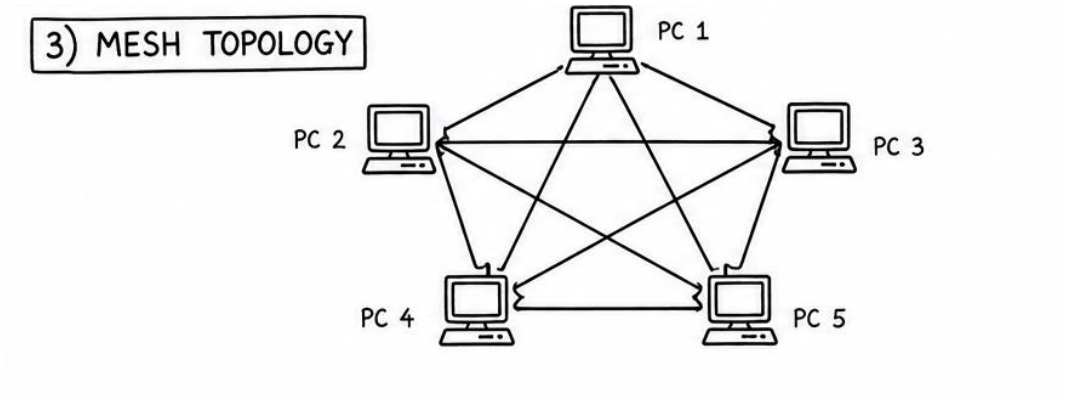
Bus Topologies :-



In a small streaming setup, multiple devices can share one network connection. This is similar to a bus topology where all devices use a single communication cable.

Bus Topology: - Suitable for small, temporary, or low-cost networks.

Mesh Topologies :-



Cloud services use multiple network paths between servers. If one path fails, data automatically travels through another path, making the network highly reliable.

Mesh Topology:- Best for cloud services, data centers, and the Internet because it provides high reliability and fault tolerance.

2.Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

Ans. :-

1. Star Topology :-

Advantages:-

- Easy to install and manage.
- If one device fails, the rest of the network continues to work.
- High network performance because each device has a dedicated connection.
- Easy to detect and fix network problems.

Disadvantages:-

- Requires more network cable.
- Installation cost is higher.
- If the central switch or hub fails, the entire network stops working.
- Depends completely on the central device.

Best Use Case: Home networks, schools, offices, and colleges.

Example App: Instagram – Users connect to Instagram's central servers to upload photos, videos, and messages.

2. Bus Topology :-

Advantages:-

- Low installation cost.
- Uses less cable than other topologies.
- Easy to set up for small networks.
- Suitable for temporary or small office networks.

Disadvantages:-

- If the backbone cable fails, the whole network stops.
- Performance becomes slow as more devices are added.
- Difficult to find faults.
- Limited cable length and number of devices.

Best Use Case: Small offices, computer labs, and temporary LANs.

Example App: Zomato – A small restaurant's local ordering or billing network can work similarly to a shared communication line.

3. Ring Topology :-

Advantages:-

- Data flows in one direction, reducing collisions.
- Every device gets equal access to the network.
- Performs well under heavy traffic.
- Organized data transmission.

Disadvantages:-

- Failure of one device can affect the entire network.
- Adding or removing devices is difficult.
- Troubleshooting is time-consuming.
- Maintenance cost is higher.

Best Use Case: Industrial systems, manufacturing plants, and banking networks.

Example App: Paytm – Payment requests are processed in an organized and secure sequence, similar to controlled data flow in a ring topology.

4. Mesh Topology :-

Advantages:-

- Very reliable because multiple communication paths are available.
- If one connection fails, data uses another path.
- High security and fault tolerance.
- Excellent network availability.

Disadvantages:-

- Very expensive to install.
- Requires a large number of cables.
- Complex configuration and maintenance.
- Needs more networking hardware.

Best Use Case: Cloud computing, data centers, military networks, and the Internet backbone.

Example App: Google Drive – Uses multiple server connections so users can access files even if one network path fails.

3. Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

Ans. :-

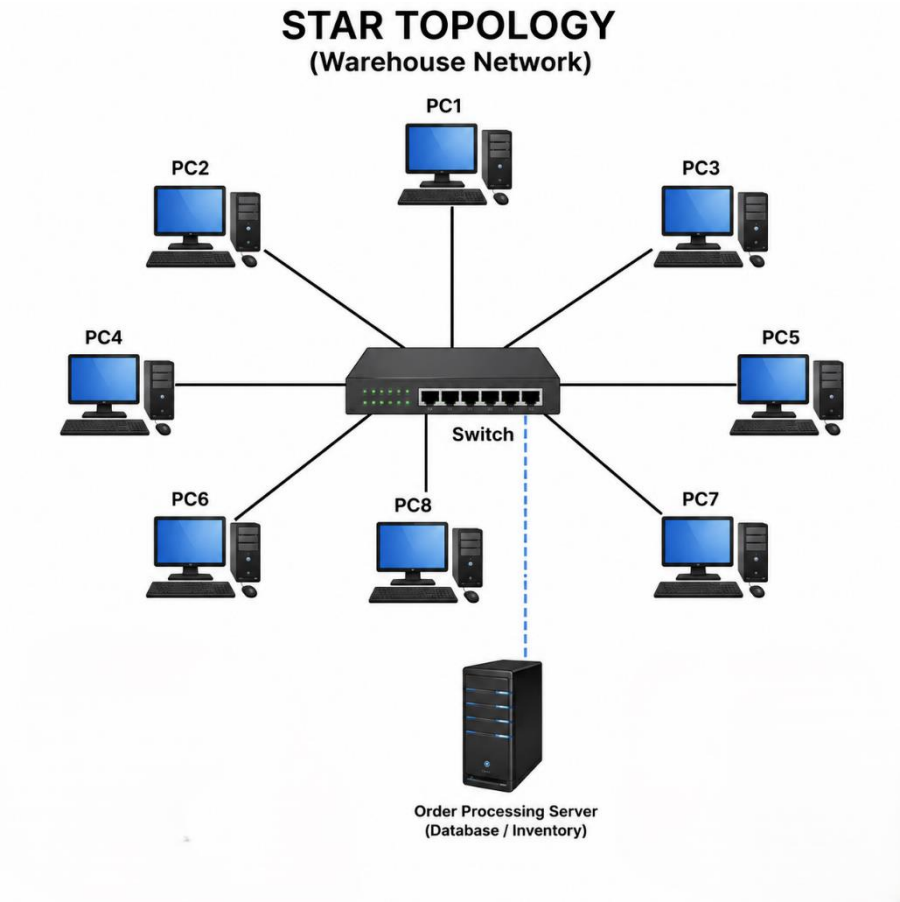
Suitable Network Topology: Mesh Topology :-

If I were designing a multiplayer mobile game like PUBG or Free Fire, I would choose a Mesh Topology for the game servers. It provides multiple communication paths, allowing data to travel through the fastest route and reducing game lag. If one server or network connection fails, another path is used automatically, making the game highly reliable and available. Although mesh topology is more expensive to install and maintain, its high speed, fault tolerance, and reliable performance make it the best choice for large-scale online multiplayer games.

**4. Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventor y updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.

Hint: Consider factors like cost, ease of troubleshooting, and future expansion.**

Ans. :-



A Star Topology is the best choice for a small Flipkart-style warehouse because it provides fast communication, high reliability, and easy troubleshooting. It is cost-effective for connecting 8 computers and allows new devices to be added easily in the future. This ensures smooth order processing, accurate inventory updates, and uninterrupted warehouse operations.